TOURISM EXPERIENCE ANALYTICS
FINAL PROJECT REPORT (ENHANCED SUBMISSION)

1. EXECUTIVE SUMMARY

This project builds a complete analytics system for tourism platforms with three integrated AI capabilities:

1. Rating prediction (regression)
2. Visit mode prediction (classification)
3. Personalized attraction recommendation (collaborative filtering)

The solution includes data preprocessing, EDA, visual analytics, ML model training/evaluation, and an interactive Streamlit application for business-facing usage.

2. PROBLEM STATEMENT AND BUSINESS OBJECTIVES

Tourism businesses need to improve user experience and retention by understanding behavior and delivering personalization at scale. This project addresses:

- Predict likely attraction rating before a trip or campaign launch.
- Classify visitor type (Business, Family, Friends, Couples, Solo) for targeted offers.
- Recommend attractions aligned with user history and similar-user patterns.

3. DATASET OVERVIEW

Integrated tourism dataset after consolidation:

- Total transactions: 52,930
- Unique users: 33,530
- Unique attractions: 30
- Time range: 2013 to 2022
- Average rating: 4.158 / 5

Core fields used:

- Behavioral: VisitYear, VisitMonth, VisitModeId, Rating
- Location: ContinentId, RegionId, CountryId, CityId
- Attraction: AttractionId, AttractionTypeId, AttractionCityId, Attraction

4. DATA CLEANING AND PREPROCESSING

Processing was performed using a reproducible Python pipeline.

4.1 CLEANING STEPS

- Filled missing categorical location labels (CityName, Country, Region, Continent) with Unknown.
- Enforced numeric consistency for ID columns and temporal fields.
- Validated and clipped ratings to the valid 1-5 range.
- Exported final cleaned dataset to data/processed/cleaned_tourism.csv.

4.2 DATA QUALITY OUTCOME

- Missing values in modeling features: 0
- Cleaned dataset shape: (52930, 21)

- Ready for EDA, ML training, and app deployment.

5. EXPLORATORY DATA ANALYSIS (EDA)
Visual outputs are saved in reports/figures/.

5.1 VISIT MODE DISTRIBUTION

Visit Mode	Transactions	Share
Couples	21,620	40.85%
Family	15,217	28.75%
Friends	10,945	20.68%
Solo	4,525	8.55%
Business	623	1.18%

5.2 TOP COUNTRIES BY TRANSACTION VOLUME

Country	Transactions
Australia	13,322
United Kingdom	6,722
United States	6,261
Indonesia	4,842
Singapore	2,807
India	2,543
Malaysia	1,581
Canada	1,486
New Zealand	1,479
Netherlands	859

5.3 MOST VISITED ATTRACTIONS

Attraction	Visits
Sacred Monkey Forest Sanctuary	13,198
Waterbom Bali	6,429
Tegalalang Rice Terrace	5,815
Uluwatu Temple	3,359
Tanah Lot Temple	3,352
Sanur Beach	3,044
Seminyak Beach	2,914
Kuta Beach - Bali	2,765
Merapi Volcano	2,235
Tegenungan Waterfall	2,190

5.4 AVERAGE RATING BY VISIT MODE

Visit Mode	Avg Rating
Business	4.313

Family	4.219
Friends	4.174
Couples	4.117
Solo	4.088

5.5 EDA INTERPRETATION
- Tourism demand is concentrated in leisure segments (Couples, Family, Friends).
- Rating levels are generally high, indicating positive user sentiment.
- Country-level concentration suggests opportunities for geo-targeted marketing and partnerships.

6. FEATURE ENGINEERING AND MODELING STRATEGY

6.1 REGRESSION TASK
Target: Rating
- Candidate models: RandomForestRegressor, GradientBoostingRegressor
- Selected by best RMSE on hold-out test split

6.2 CLASSIFICATION TASK
Target: VisitModeId
- Candidate models: RandomForestClassifier, GradientBoostingClassifier
- Selected by weighted F1 on hold-out test split
- Leakage prevention applied (target VisitModeId is not used as input feature)

6.3 RECOMMENDATION TASK
- Method: User-based collaborative filtering using KNN with cosine similarity
- Data structure: user-item matrix (UserId x AttractionId) built from ratings
- Output: top-N unseen attractions from nearest user neighbors

7. MODEL EVALUATION RESULTS

7.1 REGRESSION (BEST MODEL: GRADIENTBOOSTINGREGRESSOR)
- MAE: 0.7165
- RMSE: 0.9136
- R2: 0.1137

7.2 CLASSIFICATION (BEST MODEL: RANDOMFORESTCLASSIFIER)
- Accuracy: 0.4930
- Precision (weighted): 0.4942
- Recall (weighted): 0.4930
- F1 (weighted): 0.4411

7.3 RECOMMENDATION
- Users in matrix: 33,530
- Attractions in matrix: 30
- HitRate@5: 0.0032

7.4 PERFORMANCE DISCUSSION

- Regression captures moderate predictive signal, suitable for directional satisfaction forecasting.
- Classification performance indicates useful segmentation potential but can improve with richer behavioral features.
- Recommendation hit rate is low due to sparse user-level interactions, a common issue in implicit tourism logs.

8. BUSINESS INSIGHTS AND RECOMMENDATIONS

- Prioritize Couples/Family-focused campaign bundles because these segments dominate volume.
- Use rating prediction to proactively identify lower-satisfaction attraction-context combinations.
- Use visit mode classification to personalize messaging, package design, and amenity recommendations.
- Improve recommendation quality with additional context features (trip duration, budget, recency, dwell time).

9. STREAMLIT APPLICATION

Delivered app features:

- Executive dashboard (distribution, countries, top attractions)
- Rating prediction form
- Visit mode prediction form
- Personalized recommendation engine
- Model performance panel for transparent evaluation

File: app/streamlit_app.py

10. LIMITATIONS AND FUTURE WORK

Current limitations:

- No explicit temporal sequence modeling by user session.
- Sparse interactions reduce recommendation precision.
- Features are primarily structured IDs; semantic attraction metadata is limited.

Next improvements:

- Hybrid recommender (collaborative + content-based)
- More robust feature set (price sensitivity, seasonality intensity, travel cohort)
- Hyperparameter tuning with cross-validation and model calibration
- Better cold-start strategy for new users and new attractions

11. CONCLUSION

This project successfully implements the complete assignment scope with reproducible scripts, explainable analytics, machine learning models, and deployable UI. The output is actionable for tourism platforms and presentation-ready for academic evaluation.

12. REPRODUCIBILITY AND DELIVERABLES

- Cleaned data: data/processed/cleaned_tourism.csv
- Models: models/*.pkl
- Metrics: reports/metrics.json
- EDA charts: reports/figures/*.png
- Streamlit app: app/streamlit_app.py
- Pipeline runner: run_pipeline.py